NUH REGAİB ÜNSAL

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Education and Awards

Kocaeli University, Information Systems Engineering **2023 – Present**

- 4th year student
- Coursework in software development, databases, web technologies and cloud computing

Sabancı Mesleki ve Teknik Anadolu Lisesi, Kocaeli **2019 – 2023**

- Information Technologies Field

Experience

Software Engineering Intern, i2i Systems — YTÜ Davutpaşa Technopark, İstanbul **Jun – Jul 2026**

- Joined the i2i Academy program (29 June – 29 July 2026) and worked through 18 individual projects, covering backend, cloud, data engineering, testing and AI.
- **CryptoPal —** built a crypto trading platform as a Spring Boot modular monolith. Kept PostgreSQL as the only source of truth for balances and trades, and used Redis only for sessions and the 15-second price updates.
- **VoltWise —** sent simulated smart-home sensor data through Kafka and kept each home's live state in Apache Ignite. Homes going over 80% and 100% of their energy quota are flagged, and an LLM writes saving advice for them that is sent by e-mail.
- Wrote a Python ETL pipeline that scrapes pages, cleans the data with pandas and upserts it into PostgreSQL, so running it again does not duplicate rows.
- Set up a GitHub Actions pipeline that runs a pytest unit test and a headless Selenium test, and only deploys if both pass.
- The rest of the projects covered REST API design with Swagger, Oracle PL/SQL, Redis caching, Docker, GCP virtual machines, OpenCV plate recognition and sentiment analysis.

University and Personal Projects **2023 – Present**

- **Dispatcher API Gateway (team project) —** put three Flask services on a private Docker network so they can only be reached through the gateway. The gateway checks the JWT, limits requests per IP, and sends metrics to Prometheus and Grafana.
- **Time-Series Anomaly Detection (team project) —** compared a probabilistic automaton (PAA/SAX), which is easier to explain than a deep model, with 1D-CNN and LSTM models on the SKAB and BATADAL datasets. Fitted all preprocessing on the training split only, so nothing leaked from the test data.
- **TechStore E-Commerce Platform —** Database Management Systems course project. Designed a nine-table schema and built the store on it with PHP and MySQL, with hashed passwords and separate seller and customer roles.
- **Autonomous Image Grouping —** grouped images by similarity using MobileNetV2 features and K-Means, and picked the number of clusters with the elbow method. Ran it as Flask services in Docker.

Skills and Interests

- **Programming Languages:** Java, Python, C#, PHP, JavaScript / TypeScript, SQL (PostgreSQL, Oracle PL/SQL, MySQL)
- **Frameworks and Tools:** Spring Boot, Flask, React, Docker & Docker Compose, Apache Kafka, Redis, MongoDB, Apache Ignite, Prometheus & Grafana, Git & GitHub Actions, JUnit, pytest, Selenium, pandas, TensorFlow
- **Web Technologies:** HTML, CSS, REST APIs, Swagger / OpenAPI
- **Languages:** Turkish – Native; English – Intermediate